MILITARY SPECIFICATION

GENERATOR, TARGET, SIMULATED 11741100

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the General requirements for the Simulated Target Generator (STG).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction, M18 Manufacturing Process and Disposition of
MIL-C-48411/lA	Circuit Card Assembly: Power Control
MIL-C-48411/2A	Circuit Card Assembly: Control Panel 1
MIL-C-48411/3A	Circuit Card Assembly: Control Panel 2
MIL-c-48411/4A	Circuit Card Assembly: Accessory
MIL-C-48411/5A	Circuit Card Assembly: Timing 1
MIL-c-48411/6A	Circuit Card Assembly: Timing 2
MIL-C-48411/7A	Circuit Card Assembly: Timing 3
MIL-C-48411/8A	Circuit Card Assembly: Data Selector 1
MIL-C-48411/9A	Circuit Card Assembly: Data Selector 2
MIL-C-48411/10A	Circuit Card Assembly: Computer 1
MIL-C-48411/11A	Circuit Card Assembly: Computer 2
MIL-C-48411/12A	Circuit Card Assembly: Computer 3
MIL-C-48411/13A	Circuit Card Assembly: Computer 4
MIL-C-48411/14A	Circuit Card Assembly: Computer 5
MIL-C-48411/15A	Circuit Card Assembly: Inerface
MIL-C-48411/16A	Circuit Card Assembly: Coefficient 1
MIL-C-48411/17A	Circuit Card Assembly: Coefficient 2
MIL-C-48411/18A	Circuit Card Assembly: Position 1

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MIL-C-48411/19A	Circuit Card Assembly: Positoin 2
MIL-C-48411/20A	Circuit Card Assembly: Posion 3
MIL-C-48411/21A	Circuit Card Assembly: Position 4
MIL-C-48411/22A	Circuit Card Assembly: Scoring 1
MIL-C-484U/23A	Circuit Card Assembly: Scoring 2
MIL-C-48411/24A	Circuit Card Assembly: Bite
MIL-C-48411/25A	Circuit Card Assembly: Scoring 3
MIL-C-48411/26A	Circuit Card Assembly: Computer 6
MIL-C-48411/27A	Circuit Card Assembly: Memory I
.MIL-C-48411/28A	Circuit Card Assembly: Memory II

STANDARDS

Military

MIL-STD-105	Sampling, Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Standards, General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

Military

11741100	Generator Target, Simulated
11746252	Circuit Card Assembly: Power Control
11746254	Circuit Card Assembly: Control Panel 1
11746256	Circuit Card Assembly: Control Panel 2
11746250	Circuit Card Assembly: Accessory
11746268	Circuit Card Assembly: Timing 1
11746270	Circuit Card Assembly: Timing 2
11746264	Circuit Card Assembly: Timing 3
11746278	Circuit Card Assembly: Data Selector 1
11746276	Circuit Card Assembly: Data Selector 2
11746262	Circuit Card Assembly: Computer 1
11746258	Circuit Card Assembly: Computer 2
11746273	Circuit Card Assembly: Computer 3
11746260	Circuit Card Assembly: Computer 4
11746266	Circuit Card Assembly: Computer 5
11746274	Circuit Card Assembly: Interface
11746280	Circuit Card Assembly: Coefficient 1
11746282	Circuit Card Assembly: Coefficientr 2
11746284	Circuit Card Assembly: Position 1
11746286	Circuit Card Assembly: Position 2
11745246	Circuit Card Assembly: Position 3
11745243	Circuit Card Assembly: Positon 4
11745208	Circuit Card Assembly: Scoring 1
11745210	Circuit Card Assembly: Scoring 2
11745212	Circuit Card Assembly: Site
11745240	Circuit Card Assembly: Scoring 3
11741043	Circuit Card Assembly: Computer 6
11745203	Circuit Card Assembly: Memory I
11745258	Circuit Card Assembly: Memory II

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PACKAGING DATA SHEET

11741100 Packaging of Generator, target Simulated

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. - The Simulated Target Generator manufactured to this specification shall be capable of producing a simulated computer controlled image. The applicable detail specifications are those referred to in Table I.

3.1.1 Major component list. - The major components shall be as specified in Table I.

TABLE I - MAJOR COMPONENTS

DETAILED SPECIFICATION	TITLE	ASSEMBLY PART NO
MIL-C-48411/1A	Circuit Card Assembly Power Control	11746252
MIL-C-48411/2A	Circuit Card Assembly Control Panel 1	11746254
MIL-C-48411/3A	Circuit Card Assembly Control Panel 2	11746256
MIL-C-48411/4A	Circuit Card Assembly Accessory	11746250
MIL-C-48411/5A	Circuit Card Assembly Timing 1	11746268
MIL-C-48411/6A	Circuit Card Assembly Timing 2	11746270
MIL-C-48411/7A	Circuit Card Assembly Timing 3	11746264
MIL-C-48411/8A	Circuit Card Assembly Data Selector 1	11746278
MIL-C-48411/9A	Circuit Card Assembly Data Selector 2	11746276
MIL-C-48411/10A	Circuit Card Assembly Computer 1	11746262
MIL-C-48411/11A	Circuit Card Assembly Computer 2	11746258
MIL-C-48411/12A	Circuit Card Assembly Computer 3	11746273
MIL-C-48411/13A	Circuit Card Assembly Computer 4	11746260
MIL-C-48411/14A	Circuit Card Assembly Computer 5	11746266
MIL-C-48411/15A	Circuit Card Assembly Interface	11746274

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TABLE I - MAJOR COMPONENTS CON'T

DETAILED SPECIFICATION	TITLE	ASSEMBLY PART NO
MIL-C-48411/16A	Circuit Card Assembly Coefficient 1	11746280
MIL-C-48411/17A	Circuit Card Assembly Coefficient 2	11746282
MIL-C-48411/18A	Circuit Card Assembly Position 1	11746284
MIL-C-48411/19A	Circuit Card Assembly Position 2	11746286
MIL-C-48411/20A	Circuit Card Assembly Position 3	11745246
MIL-C-48411/21A	Circuit Card Assembly Position 4	11745243
MIL-C-48411/22A	Circuit Card Assembly Scoring 1	11745208
MIL-C-48411/23A	Circuit Card Assembly Scoring 2	11745210
MIL-C-48411/24A	Circuit Card Assembly Bite	11745212
MIL-C-48411/25A	Circuit Card Assembly Scoring 3	11745240
MIL-C-48411/26A	Circuit Card Assembly Computer 6	11741043
MIL-C-48411/27A	Circuit Card Assembly Memory I	11745203
MIL-C-48411/28A	Circuit Card Assembly Memory II	11745258

3.1.2 Government furnished property list.- Not applicable

3.2 Environmental.

3.2.1 Temperature. - The STG shall be capable of being stored at +160°F and -67°F, and operated after being temperature stabilized at +140°F and -4°F. Subsequent to such exposure the STG shall not have been damaged or adversely affected and shall meet electrical performance requirements of this specification at standard room ambient conditions.

3.2.2 Humidity.

3.2.2.1 Operating. - The STG shall be capable of meeting the performance requirements of this specification while exposed to an atmosphere with a relative humidity (RH) of 94 ± 4 percent at a temperature of 86°F.

3.2.2.2 Non-operating. - The STG shall be capable of withstanding 240 hours of continuous exposure to a relative humidity (RH) or 94 ± 4 percent at temperature cycling of from 73°F to 140°F.

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3.2.3 Shock.- The STG shall be capable fo with standing three 10G, 11 milliseconds shock in each of it nutually perpendicular axis for a total of 18 shocks.

3.2.4 Vibration. - The STG shall be capable of withstanding a vibration having a double amplitude of 0.076 inches and acceleration “levels of from 2 to 4G applied logarithmically over the 5 to 500 Hz frequency range for a 15 minute time duration.

3.2.5 Salt fog. - The STG shall be capable of withstanding forty-eight (48) hours of continuous exposure to a 5% salt solution With a specific gravity of 1.027 to 1.034, a PH of 6.5 to 7.2 and a temperature of 95°F.

3.2.6 Sand and Dust. - The STG shall be capable of operation after twenty-four (24) hours of exposure to wind driven sand and dust.

3.2.7 Rain. - The STG shall be capable of operation after exposure to a 2 inches per hour rain with 20 mph driving rain. The water temperature shall be between 60°F and 90°F.

3.2.8 Altitude. - The STG shall be capable of operating during exposure at an altitude of 10,000 ft. and after exposure at an altitude of 50,000 ft.

3.2.9 Fungus. - The STG shall be capable of withstanding a twenty-eight (28) day fungus exposure to a composite spore suspension.

3.3 Design and construction.

3.3.1 Production drawings. - The STG shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on the applicable specification and assembly drawing.

3.3.2 Standard of manufacture.- The STG shall be manufactured to meet the requirements of applicable drawing specified in Table I and MIL-F-13926 and MIL-C-46357.

3.3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Soldering of electrical connections

3.3.2.2 MIL-C-46357.

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes

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3.4 Preproduction sample. - Preproduction samples shall be given qualification test prior to regular production to demonstrate the adequacy and suitability of the contractor's processes and procedures in achieving the performance that is inherent in the design. (See 6.2)

3.5 Electrical.

3.5.1 Continuity. - The STG shall demonstrate continuity in accordance with all the wiring requirements as cited on the drawings and wiring list constituting part of the Technical Data Package.

3.6 Performance. - The performance requirement as specified herein shall be met on every assembly at standard ambient-conditions prior to Government acceptance.

3.6.1 Power requirements. - The STG console shall normally operate from a 28.0 VDC power source. through an on/off relay network. The STG console shall be capable of operating between 24.0 and 30.0 VDC with a visual indicator for under 24.0 VDC.

3.6.1.1 Digital output. - The "O" or false-output level shall be between 0.00 and 0.40 Vdc. The "1" or true output level shall be between +2.40 and ±5.00 Vdc.

3.6.2 STG/VADS Input interface indicators.- The STG console shall provide a means of monitoring the following VADS indicators:

a. Radar Mode
b. Manual Mode
c. Radiate
d. ATD (Acquisition Time Delay)
e. Ready to Fire (RFT)
f. F1/F2

3.6.2.1 Firing.- The STG shall provide an indicator for VADS FIRING A split lighted push button switch shall be used to indicate firing for two cannon bursts.

3.6.3 STG/VADS output Interface.

3.6.3.1 Radar Ready.- The STG console shall provide a 28 Vdc radar ready signal to the VADS.

3.6.3.2 + 10 VDC Reference.- The STG console shall provide a +10 Vdc reference voltage to the VADS.

3.6.3.3 -10 Vdc Reference. - STG console shall provide a -10 Vdc reference voltage to VADS.

3.6.3.4 +20 VDC Reference.- The STG console shall provide a +20 Vdc Reference voltage to VADS.

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3.6.3.5 Radar lock-on. - The STG console shall provide the VADS with a 0.0 VDC lock-on voltage and a 15.0 VDC for no lock-on.

3.6.3.6 Range. - The STG console shall provide the VADS with an analog voltage proportional to the range or the simulated target.

3.6.3.7 Range rate. - The STG console shall provide the VADS with an analog voltage proportional to the rage rate of the simulated target.

3.6.3.8 ECM (Electronic countermeasure).- The STG console shall provide the VADS with an ECM logic signal.

3.6.4 STG tracking inputs.-

3.6.4.1 Boresight switch. - The STG shall provide a boresight switch for use in initial target boresighting and azimuth turret limitations control.

3.6.4.2 Boresight box. - The STG shall provide a means of manually positioning the aircraft image in both azimuth and elevation by means of control potentiometers in the STG boresight box.

3.6.4.3 Tachometer. - The STG shall accept the VADS turret signal and convert this input into the appropriate weapon position.

3.6.4.4 Elevation pot. - The STG shall provide a means of obtaining VADS turret elevation information through a elevation pot and convert this information into the appropriate weapon position.

3.6.5 STG Console scoring.

3.6.5.1 Digital readouts. - The STG shall provide an integrated value of tracking error in Mils in both azimuth and elevation for two cannon firings per flight path. This data shall be stored upon each firing and presented on two sets of digital readouts for azimuth and elevation.

3.6.5.1.1 Self test. - The digital indicators shall provide a self test feature.

3.6.5.1.2 Fire 1 error/Fire 2 error. - A fire 1 error/fire 2 error split lighted push button switch shall be used to recall stored error information.

3.6.5.2 Tracking Error meter (Digital readout relation).- The STG console shall provide a tracking error meter which will present to the STG operator real time gunner tracking error.

3.6.5.1 Meter boresight. The STG console shall provide independent front panel adjustments of the horizontal and vertical crosshairs of the tracking error meter.

3.6.5.2.2 Reticle monitoring. - The STG shall be capable of monitoring the position of the VADS reticle in both azimuth and elevation.

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3.6.5.2.3 Reticle switch. - The STG console shall provide a momentary switch to disengage the reticle data from the STG scoring circuitry.

3.6.6 Audio. - The STG console shall provide a communication link such that the STG operator and VADS gunner shall have audio contact.

3.6.6.1 Hit tone. - The STG shall provide the operator and the gunner with a simulated tone.

3.6.7 Built-in test. - The STG console shall provide a built-in capability for fault isolation.

3.6.8 Flight paths. - The STG shall provide the capability of dynamically computing 100 pre-programmed aircraft flight paths.

3.6.9 STG Console/VIG Interface.

3.6.9.1 Target attitude. - The STG console shall provide the VIG (Visual Image Generator) with six co-ordinated signals indicative of the attitude of the simulated aircraft target.

3.6.9.2 Size. - The STG console shall provide the VIG with a signal indicative of target size.

3.6.9.3 +10 Vdc Reference. - The STG console shall provide the VIG with a +10 VDC reference signal.

3.6.9.4 Target blanking. - The STG shall provide the VIG with a signal to indicate a smoke/no smoke condition.

3.7 Reliability run-in. - The STG shall be subjected to a 50 hour run-in period (excluding downtime) which includes the operation at temperature extremes. The 50 hour run-in period shall start after the STG has satisfactorily met the performance requirements of this specification. The time required to conduct the vibration test shall not be included as part of the run-in period. The STG shall continuously meet the requirement as specified in 4.4.4 for the entire 50 hour run-in period.

3.7.1 Vibration. - The STG shall not be damaged after being subjected to simple harmonic motion for a period of 30 minutes.

3.7.2 Temperature extremes.- The STG shall be capable of operating without damage for 4 hours at -4°F and for 4 hours at 125°F.

3.7.3 Failures during reliability run-in.- Replacement of defective components and correction of workmanship defects may be made during run-in under the following conditions:

a. Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the reliability run-in environmental requirements of the STG .

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b. Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement for any failure occurring during a temperature extreme.

c. After a correction has been accomplished, at least one (1) hour of operation with a minimum of three (3) tests to the part, module, component or subassembly requirements shall be successfully accomplished at the temperature at which the failure occurred.

3.8 Reliability demonstration. - The Mean Time Between Failure (MTBF) of the STG console shall be 300-hours minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- Unless otherwise specified in the contract, the supplier is responsible for the performance of all test and inspection requirements specified herein. Except as "otherwise specified in the contract the supplier will specify facilities suitable for the performance of the test and inspection, unless disapproved by the Government. The Government reserves the right to Perform any of the tests and inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.1.2 Special tests and examinations.

4.1.2.1 First article (initial production) approval.- The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 3 assemblies manufactured in the same manner using the same materials, equipment, processes, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The assemblies shall meet the requirements and tests of Table II.

4.1.2.1.1 Government testing. - When the Government is responsible for conducting first-article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials,

4.1.2.1.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.1.2.2 Control tests. - Three assemblies shall be selected at random as a control simple from each 50 produced. The control samples shall meet the requirements and tests of Table II. (See 6.3)

4.1.2.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.2.2).

TABLE II

TEST AND REQUIREMENTS

TEST	REQUIREMENT	TEST PROCEDURES
Performance	3.6	4.3.12
Temperature	3.2.1.1.1	4.3.1
Humidity	3.2.1.1.2	4.3.2
Shock	3.2.1.1.3	4.3.3
Vibration	3.2.1.1.4	4.3.4
Salt Fog	3.2.1.1.5	4.3.5
Sand and Dust	3.2.1.1.6	4.3.6
Rain	3.2.1.1.7	4.3.7
Altitude	3.2.1.1.8	4.3.8
Fungus	3.2.1.1.9	4.3.9
Design and Construction	3.3	Visual

4.2 Quality conformance inspections.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” as specified in MIL-STD-105.

4.2.2 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.2.3 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor’s delivery schedule.

4.2.3.1 Contractor designed equipment.- The contractor shall provide inspection equipment compatible with the “Test Methods and Procedures” specified in the detail specification and with the “Component Inspection” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the

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test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, material~, dimensions and tolerance used in the design test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerances limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment,

4.2.3.2 Test equipment.- Test equipment used to test assemblies for verification of Section 3 electrical requirements shall be as specified in Table III.

4.2.4 Examinations.- Unless otherwise specified, components and subassemblies shall be inspected in accorcancw with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations to verify requirements of 3.3 shall be performed. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.5 Functional tests.- Functional requirements shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and test shall be conducted at the following standard ambient conditions:

a. Temperature	+40°F to +90°F
b. Relative humidity	90 percent RH max.
c. Pressure	28 to 32 inches of Hg

4.3 Environmental Test Methods and Procedures.

4.3.1 Temperature storage.

a. High temperature.- This test shall be conducted in accordance with Procedure I, Method 501 of MIL-STD-81OB.

b. Low temperature. - This test shall be conducted in accordance with Procedure I, Method 502 of MIL-STD-810B.

4.3.2 Humidity. - The STG shall be subjected to the humidity test as outlined in MIL-STD-81O, Method 50.7, Procedure II, except that performance test to be conducted during the exposure shall be conducted during the lst, 3rd and 5th cycles; subsequently the STG shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.3 Shock. - The STG shall be subjected to three 10G, II millisecond duration, shocks in each of the three mutually perpendicular directions for a total of 18 shocks as outlined in MIL-STD-810, Procedure I, Subsequent to the shock test, the STG shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

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4.3.4 Vibration. - The STG shall be subjected to the vibration as specified in MIL-STD-810, Method 514, Procedure VIII except that the displacement and frequency shall be as follows:

GRAVITY	FREQUENCY (Hz)
0.076 in. double amplitude	5-23
2G	23-40
4G	40-500

Subsequent to the vibration test, the STG shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.5 Salt fog. - The STG shall be subjected to the salt fog test as specified in MIL-STD-810, Method 509. Subsequent to the exposure the STG shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.6 Sand and Dust. - The STG shall be subjected to the sand and dust test as specified in MIL-STD-810, Method 510. Subsequent to the exposure the STG shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.7 Rain. - The STG shall be subjected to the rain test as specified in MIL-STD-810, Method 506. The STG will not be operated during the test, but subsequently it shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.8 Altitude. - The STG shall be subjected to the altitude as specified in MIL-STD-810, Method 500, Procedure I. Subsequent to the exposure, the STG shall not have been damaged to adversely affected and shall meet the performance requirements of this specification.

4.3.10 Test Equipment.

ITEM	DESCRIPTION
Power Supply	Range 0 -50 VDC, 20A 115 VAC 400 HZ lAMP Accuracy ±1.0% Sensitivity 20K oHM/vDc
Voltmeter, Digital	Range 0-500 VDC 0-250 VDC Accuracy and Resolution 0.1% or better across the range. Input Impedance - 10 MEG or better for all voltage scales.
Oscilloscope	Rise Time for scope and asso- ciated measuring probe of 50 nanoseconds or greater Sweep Period - 50 nanoseconds per centimeter or greater

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ITEM	DESCRIPTION
Oscilloscope (Con't)	Input Resistance of greater than 8 Megohms. Input Capacitance of less than 25 picafarads.
Electronic Counter	Range DC coupled, O to 50 MHz AC coupled 25 Hz to 50 MHz Gate Time 1 us to 10 sec. in decade steps. Accuracy ± 1 count ± time base accuracy.
Synchro	Part No. 11740911
Box Boresight	Part No. 11745276
Cable Main Interface (w1/w34	Part No. 11745270

4.3.11 Electrical.

4.3.11.1 Continuity. - The STG shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.5.1.

4.3.12 Performance. - Unless otherwise specified, the performance tests as cited herein shall be satisfactorily met at standard ambient conditions standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mecury.
c. Relative humidity (RH) up to 90%.

4.3.12.1 Test Set-up. - The input voltage shall be 28.0 ± 0.1 VDC and the initial test switch positions shall be as follows:

SWITCH	POSITIONS
a. Power	On
b. Run Select	52
c. Run/Hold	Hold
d. Simulator/Tester	Simulator

4.3.12.1.1 Testing. - Subsequent to each power on/test cycle the reset switch shall be depressed and the boresight indicator depressed and released such that the indicator light is extinguished.

4.3.12.1.2 Digital output. - The "O" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and 5.45.

4.3.12.1.3 Power requirements. - In order to determine compliance with paragraph 3.6.1, there shall be continuity between pins J3= and J4A and between J3m and J4A when no power is applied. When power is applied and the on/off switch in the "off" position the red indicator shall glow. With With

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the switch in the “on” position the green indicator shall glow.

4.3.12.1.4 +28.0 VDC and internal power supply.- When +28.0 ±0.l VDC is applied to pin J4A with J4C as a return the voltages for the power supplies shall be capable of being adjusted. The values as shown below.

PS1	To + 15.000 ± 0.050 VDC
PS2	To - 15.000 ± 0.050 VDC
PS3	To + 5.000 ± O.050 VDC
PS4	To - 28.000 ± 0.100 VDC
PS5	To + 115.0 VAC

4.3.12.1.5 +24.0 and +30.0 VDC input voltage.- When input voltages of 24.0 -00 and 30.0 +0.0 VDC are applied and the adjustment for 4.3.12.2 unchanged, the respective power supply voltages shall be as shown below.

Ps1	+15.000 ± 0.250 VDC
PS2	-15.000 ± 0.250 VDC
PS3	+5.000 ± 0.250 VDC
PS4	+28.000 ± 0.300 VDC
PS5	115.0 ± 2 VAC

4.3.12.1.6 Low voltage indicator.- When the input voltage is 23.8 ± 0.2 VDC the power “on” portion of the on/off switch shall blink.

4.3.12.2 STG/VADS input interface indicators.

4.3.12.2.1 Radar mode. - With STG power on and no signal input on connector J1 or J2, observe that the radar mode indicator lights.

4.3.12.2.2. Manual mode. - With +28.0 ± 0.5 VDC applied to J1-f J1-s as the return, the manual lamp shall be on and the radar mode light shall extinguished.

4.3.12.2.3 Radiate. With +28.O ± O.5 VDC applied to J1-W and J1-s as the return, the radiate indicator shall light.

4.3.12.2.4 ATD on. - With the voltage applied as in 4.3.12.2.3, place both crosshairs of meter M4 in the 15 mil reticle by using the crosshair adjust switch and console boresight potentiometer. The ATD and lock-on indicators shall light.

4.3.12.2.4.1 ATD off. - Subsequent to 4.3.12.2.4, with the +28.0 +0.5 VDC applied to J1-b and J1-s as the return, the ATD indicator shall not light.

4.3.12.2.5 RTF off.- With +28.0 ± 0.5 VDC applied to J1-v and J1-s as the return, the RTF indicator is out.

+28.0 ± 0.5 VDC applied to J1-q, the RTF indicator shall light.

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4.3.12.2.6 F1/F2. - Subsequent to 4.3.12.2.5.1 and with a +28.0 ±0.5 VDC pulse of 0.2 seconds duration applied to J1 w and J1 s as a return, the F1 portion of F1/F2 shall light. Apply a second pulse and the F2 portion shall light. Depress the reset switch and both light shall be extinguished.

4.3.12.2.6.1 F1/F2 blinking. - Subsequent to 4.3.12.2.6 with the +28.0 VDC removed from J1 q, the RTF lamp shall be extinguished and F1/F2 shall blink. Depress the reset switch and the lights shall be extinguished.

4.3.12.3 STG/VAD output interface.

4.3.12.3.1 Radar ready. - With power applied, connect a DVM across J1 Y and JI s. Using J1 s as ground. The DVM shall read +28.0 ± 0.5 VDC.

4.3.12.3.2 +10 VDC Reference. - With power applied, connect a DVM across J1-FF and J1-dd. Using J1-DD as ground, the DVM shall read +10.0 ± 0.1 VDc.

4.3.12.3.3 -10 VDC Reference. - With power applied, connect a DVM across J1-GG and J1-DD as ground, the DVM shall read -10.0 +o.l VDc.

4.3.12.3.4 +20 VDC Reference. - With power applied, connect a DVM across J1-L and J1-M. Using J1-M as ground, the DVM shall read +20.0 ± O.3 VDC.l

4.3.12.3.5 Radar lock-on. - With power applied, connect a DVM across J1-AA and J1-M. Using J1-M as ground, the DVM shall read 15.0 ± O.3 VDC. Repeat 4.3.12.4 and the DVM shall read 0.0 ± O.5 VDC and the lock-on shall light. Readjust both crosshairs of the tracking error meter to outside the 60 mil reticle and the DVM shall read 15.0 ± O.3 VDC after a 1.0 ± O.2 sec delay.

4.3.12.3.6 Range. With power applied, connect a DVM across J1-K and J1-L. Using J1-L as ground. The DVM shall read 6.4+0.1 VDC. Run flight 52 until range meter M2 indicates a range of 1500+200 meters. Depress hold switch. With DVM measure and record voltage between J3-R and J3-A (with J3-A as ground). Repeat 4.3.12.2.4 and the voltage at J1-K is 1.6+0.02 times the value of J3-R.

4.3.12.3.7 Range Rate. - With power applied, connect a DVM across J1-CC and J1-L. Using J1-L as ground, the DVM shall read -6.4 ± 0.1 VDC. Set run select switch to run 04 and with the console depress the reset switch. Repeat 4.3.12.2.4 and the DVM shall read -3.24 ± 0.20 VDC.

4.3.12.3.8 ECM. - With power applied, connect an oscilloscope across J1-U and J1-s. Using J1-S as ground, depress and hold indicator ECM and observe on the oscilloscope logic “1” and that the ECM indicator lights. Release ECM indicator and observe blinking of the light and the appearance of a 7.5 ± 1.0 H2 square wave. The square wave shall switch alternately from logic “l” to logic “0”.

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4.3.12.4 STG Tracking Inputs.

4.3.12.4.1 Boresight switch. - With run selector switch in run 52. Apply console power and set run/hold switch in hold. Reset and depress and release boresight switch. Depress boresight switch and the indicator shall light and that altitude meter indicates 0+20 meters. Release boresight switch and indicator light shall extinguish and meter reads 90(&50 meters. Depress boresight switch and measure the voltage across J3N and J3-A with DVM. Apply +10.0 ± 0.1 VDC differential signal across J1 p and J1-n, with J1-n as goung. The DVM reading shall not be affected by the 10 volt signal.

4.3.12.4.2 Boresight box (Azimuth). - With STG.console energized as in 4.3.12.4.1, connect P2 of cable assembly W1/W34 to console connector J1. Attach STG boresight box to W1/W34 P1O. Connect a DVM across J3 N and J3 A, using J3-A as ground. Rotate azimuth potentiometer on boresight box fully clockwise and then counterclockwise noting the DVM reading at each extreme. The DVM shall read a total range of 3.4 ± 0.5 VDC symmetrically 0.0 ± 0.5 VDC.

4.3.12.4.3 Boresight box (Elevation).- With console connected as above except pin J3 N is replaced by J3 P and the elevation potentiometer is rotate. The DVM shall read 8.0 ± 10.0 VDC symmetrically about 0.0 ± 0.5 VDC.

4.3.12.4.4 Tachometer. - Set the azimuth adjust potentiometer on the rear of the console to 1000 ± 5 OHMS with the aid of an ohmmeter. With the boresight switch depressed, apply a 8.14 ± 0.02 VDC differential signal across J1 n and J1 p, using J1 p as ground. Connect an oscilloscope to J1-T with J1 s as ground and measure logic “0”. Using a stop watch, simultaneously start the stop watch and release the boresight switch.. Stop the watch when the scope indicates logic “1”. The stop watch shall read 10+1 second and the boresight indicator shall blink. Reverse test leads on J1 n and J1 p and repeat the above test and observe the same-results. “

4.3.12.4.5 Elevation Potentiometer.- With STG energized, connect P2 of W1/w34 to console connector J1. Attach STG boresight box to P10 of W1/W34 and elevation potentiometer to W25. Connect DVM across J2-E and J2-A, with J2-A as ground. Rotate shaft of elevation pot assembly and the DVM shall read between -2 and -15+ ± 1 VDC.

4.3.12.5 STG Console Scoring.

4.3.12.5.1 Digital readout.

4.3.12.5.1.1 Self test. - Depress flight ended indicator and all digital readouts shall indicate “8”.

4.3.12.5.1.2 Fire 1 error/Fire 2 error.

4.3.12.5.1.2.1 Depress boresight switch (light on). Adjust azimuth crosshair to the left of center such that the azimuth digital readouts read 30 ± 2. Set elevation crosshair to the upper edge of the 15 mil reticle such that the elevation digital readouts read 07+2. Depress Fire l/Fire 2 error indicator such that upper portion (fire 1 error) is lit. Perform 4.3.12.4.6 move both crosshairs to the maximum extremes allowable by the pot controls. both sets of digital readouts should not change with crosshair movement (record readout indications.)

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4.3.12.5.1.2.2 Depress DS14 such that fire 2 error portion of indicator is lit. Adjust azimuth crosshair to read 07 ± 01 and elevation crosshair to read 30 ± 01. Crosshairs maybe on either side of the meter center. Perform 4.3.12.4.6.1. Move both crosshairs to the left and “upper entremes allowable by pot controls. Both sets of digital readout should not change with cross-hair movement. (record readout indications)

4.3.12.5.1.2.3 Place both crosshairs to the approximate center of the 15 mil reticle. Depress DS14 such that fire 1 error portion is lit. Readouts should indicate values noted in 4.3.12.7.1.2.1.

4.3.12.5.1.2.4 Depress DS14 such that fire 2 error portion is lit. Readouts should indicate values noted in 4.3.12.7.1.2.2.

4.3.12.7.1.2.5 Depress reset switch, move crosshairs. Readouts shall now change with movement of crosshairs.

4.3.12.5.1.3 Tracking error meter-digital readout relation.

4.3.12.5.1.3.1 Meter boresight.- Depress boresight switch light on) engage reticle switch on rear of console. Adjust AZ and EL posts on the front panal to center both crosshairs of the tracking error meter to the center of the 15 mil reticle read all zero’s on digital readouts.

4.3.12.5.1.3.2 Rotate each pot fully clockwise and counterclockwise. Crosshairs shall be outside of the 60 mil reticle at the pot extremes.

4.3.12.5.1.3.3 Release boresight switch (lightout) both sets of read-outs read 49.

4.3.12.5.1.3.4 Depress boresight switch (light on) adjust AZ and EL pots on console to left and upper edge of the 15 mil reticle. Each set of digital readouts shall read 07 ± 0l.

4.3.12.5.1.3.5 Repeat 4.3.12.7.1.3.4 with crosshairs at right and lower edge of 15 mil reticle.

4.3.12.5.1.3.6 Adjust AZ and EL pots on console to left and upper edge of 30 mil reticle. Each set of digital readouts shall read 30+01.

4.3.12.5.1.3.7 Repeat 4.3.12.7.1.3.6 for right and lower edge of 30 mil reticle.

4.3.12.5.1.3.4 Reticle monitoring. - Connect an oscilloscope probe to pin AA or J2 and the ground lend to pin s. Connect another probe to pin V of J2 with the scope in the chop mode, apply a 28 ± .5 VDC signal be-tween pins w and s of J1. The radiate, lock-on and ATD indicators shall light. The voltage At J2-Z shall be 0 ± 0.5 VDC. The voltage on J2 r shall be a logical O.

4.3.12.5.1.3.8.1 Rotate the AZ adjust pot on the console such that the azimuth crosshair is outside the 60 mil reticle the voltage on J2 r shall switch to logical 1 and the voltage on J2-Z shall go to 15.0+0.3 VDC 1.0+0.2 seconds after J2 r changes state. When this 15 volt level is reached, the lock-on and ATD indicate will go out.

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4.3.12.5.1.3.8.2 Set AZ and EL readouts to 00. Connect a synchro in accordance with appendix B, Table I. Adjust the synchro clockwise to 47 mils. Read 30 ± 02 on azimuth and elevation digital read outs. Repeat for 47 mils counter clockwise rotation.

4.3.12.5.1.3.9 Reticle switch.- With the 47 mils counterclockwise synchro input on J1, activate the reticle switch. Both sets of readouts shall read 00.

4.3.12.6 Audio

4.3.12.6.1 Connect cable W1/W34 on connector J1. Connect W1/W34 J10 to Boresight box. Connect a headset to the headset connectors on the console and boresight box by connecting yellow marked headset plug to yellow marked headset plug to yellow marked connectors on console and boresight box. Personnel wearing headsets shall have noise free communications with audio amplitude equal to normal conversational level.

4.3.12.6.2 Place AZ and EL crosshairs in center of 15 mil reticle with digital readouts at 00 or 01 by using crosshair adjust switch and AZ and EL adjust pots on console. Perform prodedure 4.4.5. Upon the application of the pulse a 2 second tone shall be heard in both headsets.

4.3.12.6.2.1 Place any crosshair to the right such as digital readout reads 09. Upon application of the pulse no tone shall be heard. Perform the same procedure with the reticle to the left of center.

4.3.12.6.2.2 Recenter azimuth crosshair and repeat 4.8.2.1 for elevatio crosshair. No tone shall be heard with the digital readouts at 09.

4.3.12.7 Built-in-test (see Appendix B)

4.3.12.8 Flight paths (see appendix A)

4.3.12.8.1 Pre-set Stops.

a. Turn console power on.
b. Set run/select switch to 00.
c. Simulator/Tester switch to simulator.
d. Run/Hold switch to Hold.
e. Depress and release boresight switch.
f. Depress and release reset switch.

Observe the following. indications

a. Simulator portion of Simulator/Tester switch lights
b. Hold portion of Run/Hold switch lights.
c. Reset switch lights when depressed and extinguishes when released.

4.3.12.8.2 Connect a DVM between pins R and A of connector J3 with pin A grounded. Measure +5.2 ± .2 VDC. The range meter M2 shall read 2080 ± 150 meters.

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4.3.12.8.3 Place Simulator/Tester switch in Tester position, and self test rotary switch to Position 1. Tester segment of switch shall light and elevation digital “readouts shall read 00.

4.3.12.8.4 Place Run/MHold switch in Run position and depress and release reset. Run segment of switch shall light and elevation digital readouts shall indicate sequential counts at 1 second intervals.

4.3.12.8.5 Repeat drove test. Monitor the digital counts with a stop watch. Start the watch upon releasing the reset switch and stop the watch when the digital indicator reads 56. The watch shall read 56 ± 02 seconds. The digital indicators shall stop counting at 56. The crossover indicator shall be lit and the flight ended indicator turn on at the count of 56.

4.3.12.8.6 Place self test rotary switch to position 8. Set Run Select switch to run 52. Place Run Hold switch in Run. Depress and release reset switch. Digital indicators shall count, stopping at 20 ± 02 counts. The crossover indicator shall light when the last count is reached. The voltage on J2 j and s with s at ground shall be O ± 0.10 VDC.

4.3.12.8.7 Place Simulator/Tester switch to the Simulator position and Run/Hold swithc in Run. Observe that the Flight Ended indicator lights when the following two indications are achieved:

a. Crossover indicator is lit.
b. Range meter reads 2600 ± 200 meters.

4.3.12.8.8 Using Flight 52 place console in run and observe change of ranges azimuth and elevation on meters. 10 ± 4 seconds after flight initiation place Run/Hold indicator in Hold. Movement of all meters shall stop.

4.3.12.9 STG Console/VIG Interface

4.3.12.9.1 Target attitude.- Measure the following voltages on connector J3 using DVM. For all measurements J3-A is ground. One set of measurements is made with the Simulator/Tester switch in Simulator and the other set with the switch in Tester, Tolerance for all readings is ± 0.4 VDC .

	SIMULATOR	TESTER
J3-D	-9.95	-4.49
J3-C	-0.98	-0.95
J3-B	0	-8.73
J3-G	-0.27	-8.78
J3-F	2.79	0.61
J3-E	9.57	4.45

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4.3.12.9.2 Size. - Connect a DVM to J3-H and J3-J with J3-A at ground rotate the image size switch through its four positions. The following logic levels shall be measured.

	J3-H	J3-J
X1	0	0
X2	1	0
X3	0	1
X4	1	1

4.3.12.9.3 +10 VDC Reference. - Connect a DVM between J3-T and J3-A with pin A at ground. Measure 10 ± 0.1 VDC.

4.3.12.9.4 Target blanking. - Connect an oscilloscope between J3-S and J3-A witn pin A at ground. Place the Smoke/Normal switch on the rear of the console in the Normal position. Measure a logical "0" on the scope.

4.3.12.9.4.1 Depress the Manual Mode and Fire 1 indicators. The Fire 1 indicator shall light and the scope shall read a logical "O". Reset the console.

4.3.12.9.4.2. Apply a +18 ± 0.1 VDC signal on J1-c with J1-s at ground. Place the Smoke/Normal switch in Smoke. Repeat 4.3.4.1. The scope shall indicate a logical "1".

4.3.12.9.4.3 Remove the +18 VDC signal. The output on J3-s shall return to logical "1".

4.3.12.10 Programmed Flight Paths.- The 100 programmed flight paths listed in Appendix A shall be verified using the pre-set stops of paragraphs 4.3.12.8.5 and 4.3.12.8.6. The following values shall be measured with the associated tolerances.

Initial azimuth	± 200 mils
Initial range	± 100 meters
Initial elevation	± 50 meters
Crossover azimuth	± 300 mils
Crossover range	± 200 meters
Crossover elevation	± 100 meters
Cross over time	± 2 counts
Final azimuth	± 200 mils
Final range	± 200 meters
Final elevation	± 50 meters
Time of flight path	± 3 counts

4.3.12.10.1 Initial measurements procedure.

a. Set run select switch to appropriate run number.
b. Run/Hold switch to Hold.
c. Simulator/Tester switch in Simulator.
d. Boresight Indicator out.
e. Place console in Simulator mode and read final position values on meters.

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4.4 Reliability Run-in. - Subsequent to the successful performance requirements of this specification the STG shall be subjected to 50 hours on Run-in time. The Run-in shall exclude any down-time necessary for repairs. It shall include vibration, followed by at least 4 hours at -4°F and at least 4 hours at +125°F.

4.4.1 Vibration run-in. - The vibration test shall be performed as specified in MIL-SID-810, Method 514.1, Procedure IX, Part 1. Subsequent to the vibration test, the STG shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.4.2 Temperature run-in. - The STG shall be placed in a chamber and the temperature gradually raised at a rate not to exceed 2°F per minute until thermally stabilized at plus 125°F. Five minutes after power is applied, test to the performance requirements of paragraph 4.4.4 shall commence. A second complete test to the requirements of paragraph 4.4.4 shall be accomplished during the middle of the four hour period and a third test to the performance requirements of paragraph 4.4.4 shall be accomplished during the last part of the hour test. Repeat the above tests at a temperature of -4°F. Subsequent to the above temperature extremes (plus 125°F and -4°F) the STG shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.4.3 Failures. - Failures during Reliability Run-in shall not be cause for rejection, but shall be corrected as outlined in 3.7.3.

4.4.4 Run-In-Tests. - The STG shall meet all the performance requirements of 4.3.12 with the exception of the following:

a. Paragraphs 4.3.12.1 through 4.3.12.1.4 shall not be tested.
b. Tests performed under paragraph 4.3.12.2.2 shall be limited to flight path number 4, 12, 15, 26, 33, 48, 52, 66, 78, 86, 95.

4.5 Reliability Demonstration. - Subsequent to the Reliability Run In, a reliability demonstration test shall be conducted as part of First Article Test to assure compliance with para 3.8 at a 70% confidence level. The system shall operate at least 4 hours during exposure to each operating environments of 3.2 and after each non-operating environment of 3.2. Additional hours shall be accumulated at 75 ± 15°F until the total test operating hours meet the accept/reject criteria of Table III. The performance requirements of para 3.6 shall be performed at least once every four hours of operation during ambient temperature exposure to the operating environments of para 3.2. The sample size shall be as a minimum the First Article Sample size. The data obtained from the Reliability Run In Test may be used as part of the Demonstration Test based upon approved by the Frankford Arsenal Reliability & Maintainability Engineering Branch, SARFA-QAF-R. The approval will consider failures in terms of being charged to reliability.

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TOTAL TEST TIME	CHARGEABLE FAILURES ACCEPT	REJECT
360	0	4 or more
730	up to 1	4or more
1090	up to 2	4 or more
1430	up to 3	4 or more

4.5.1 Data collection. - Data collected will consist of all equipment malfunctions, number of hours completed when malfunction occurred, nature and cause of malfunction, time required to correct malfunction, required corrective action, item serial number and environmental exposure.

4.5.2 Failure definition. - A failure is defined as any malfunction which renders the system inoperative or operative outside the limits of this specification shall apply:

a. If upon failure analysis (to determine the cause of the malfunction) an assignable cause (related to manufacturing process, procedure, basic design weakness, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure. However, correction to the manufacturing process or procedure (et al) shall be accomplished.

b. If upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be a “chargeable” reliability failure and as such must be considered in the above demonstration test criteria. The decisions as t ochangeable reliability failure must be co-ordinated with the procuring activity, Frankford Arsenal Reliability and Maintainability Engineering Branch, SARFA-QAF-R.

5. PREPARATION FOR DELIVERY

5.1 General. - Packaging and marking shall be in accordance with the requirements as specified in the applicable Packaging Data Sheet (PDS), AMC form 1029.

6. NOTES

6.1 Intended use. - The STG covered by this specification is a type of electronic equipment capable of controlling a simulated aircraft image against the real world as a background, ‘which is used to evaluate gunner proficiency. It has the capability of operating either remotely from the gun or in a temperature controlled shelter.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Instructions for arranging of examination, test, and approval of first article (see 4.2).

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6.3 Instruction for models and samples.

6.3.1 Instruction for models and samples.

6.3.1 Instruction for preproduction sample. - Preproduction samples shall be as specified in the contract.

6.3.2 Tests. - Any change to control tests specified in 4.1.2.2 is to be so specified in the contract.

Custodian:

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Preparing Activity:

Army - MU

Project No. 1240-A557

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METER READINGS

		AZIMUTH			RANGE			ALTITUDE			
FLIGHT TIME	PATH	INITIAL	TIME X-OVER	FINAL	INITIAL	X-OVER	FINAL	INITIAL	X-OVER	FINAL
56	0	1350	39 6381	5324	2080	585	1060	305	305	305
56	1	5050	39 6382	1078	2084	586	1059	305	305	305
56	2	5001	39 6378	1169	2041	405	971	61	61	61
56	3	1473	39 6366	5082	2016	258	920	61	61	61
56	4	4927	39 6363	1321	2016	258	920	61	61	61
56	5	1329	34 6307	5185	2672	766	1921	305	305	305
56	6	5018	33 6382	1217	2724	928	1993	610	610	610
56	7	1442	34 6237	5028	2593	409	1809	61	61	61
56	8	4958	33 6368	1375	2610	503	1834	305	305	305
56	9	1257	41 6164	5038	2120	348	865	61	61	61
56	10	1600	28 320	5165	1501	270	1431	61	61	61
56	11	1328	34 6298	4770	2656	252	1699	61	61	61
56	12	5246	36 261	1551	2844	583	1647	305	305	305
51	13	4800	25 6108	1246	2023	464	2022	305	305	305
56	14	5246	36 261	1551	2893	501	1620	610	69	69
56	15	5248	35 104	1551	2844	548	1670	305	217	411
56	16	5072	32 6270	1217	2724	836	2028	610	447	718
56	17	1474	39 6366	5082	2016	323	940	61	203	203
56	18	4958	33 6368	1375	2593	471	1829	61	249	277
56	19	1156	24 969	1424	1683	584	1339	305	305	305
56	20	5600	37 6100	1033	2122	393	1007	61	61	61
56	21	945	30 6313	5411	2519	524	2260	305	305	305
56	22	1155	24 968	1424	1683	584	1339	305	305	305
56	23	944	29 99	5411	2501	435	2240	61	61	61
56	24	5001	35 5414	5674	2041	360	1148	61	61	61
56	25	1432	29 33	485	1522	257	737	61	61	61
56	26	1600	47 6182	5321	3271	716	977	305	305	305
56	27	1600	36 6186	1052	2692	313	1617	61	61	61
50	28	5487	25 6337	1505	1922	282	1921	61	61	61
49	29	5487	26 0	1121	1922	241	1872	61	61	61
49	30	4958	24 6175	1439	2610	511	2535	305	305	305
49	31	1442	24 228	4963	2610	511	2535	305	305	305
49	32	9800	24 6106	1247	2579	542	2580	305	305	305
49	33	1600	24 297	5156	2579	542	2580	305	305	305
32	34	1660	16 273	5334	2579	730	2574	305	305	305
49	35	4800	24 6106	1247	2633	460	2592	610	108	391
33	36	5072	17 104	1329	2667	766	2652	305	305	305
32	37	5072	17 104	1312	2803	852	2532	914	483	480
37	38	1240	22 6251	4889	3523	650	2571	610	175	973
38	39	1240	22 6251	4873	3483	686	2548	305	305	305
38	40	1240	22 6251	4872	3523	634	2534	610	154	154
38	41	1240	22 6251	4873	3471	618	2530	61	61	61

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METER READINGS

FLIGHT TIME	FLIGHT PATH	AZIMUTH INITIAL	AZIMUTH TIME X-OVER	AZIMUTH FINAL	RANGE INITIAL	RANGE X-OVER	RANGE FINAL	ALTITUDE INITIAL	ALTITUDE X-OVER	ALTITUDE FINAL
33	42	5072	17 98	1328	2671	766	2649	305	305	305
34	43	5247	18 257	1331	2846	582	2634	305	305	305
34	44	5247	19 565	1331	2974	649	2621	914	392	157
33	45	4997	17 134	1404	2627	589	2603	305	305	305
33	46	1404	17 6269	4998	2627	589	2602	305	305	305
35	47	5246	18 257	1331	2831	543	2639	61	220	348
42	48	1600	21 121	4851	3274	601	2555	305	305	305
42	49	1600	21 121	4851	3261	555	2623	61	201	667
42	50	1600	21 121	4851	3261	521	2629	61	61	691
38	51	976	20 5976	5304	3145	474	2563	305	305	305
38	52	976	20 5976	5304	3216	585	2586	914	459	459
49	53	4899	25 6049	1500	2574	258	2498	61	61	61
49	54	1501	25 6346	4903	2574	257	2497	61	61	61
32	55	4800	16 6106	1239	2579	542	2479	305	305	305
38	56	1600	22 6270	4850	3261	687	2520	61	516	228
38	57	1600	20 342	5053	3274	860	2619	305	154	595
36	58	4800	20 6007	400	3274	634	2641	305	259	828
26	59	4800	15 6019	636	3261	727	2559	61	334	267
28	60	1600	16 21	5006	3261	554	2565	61	175	516
28	61	4800	15 5763	865	3261	396	2695	61	139	212
32	62	4800	16 6129	1068	2633	690	2611	610	192	535
25	63	1600	12 423	5334	2001	555	2030	61	185	365
29	64	1600	17 6022	4870	3395	580	2659	914	267	264
28	65	1600	16 127	5116	3274	835	2540	305	521	299
27	66	4800	16 6373	660	3261	576	2564	61	61	495
27	67	5424	15 5877	818	3145	589	2592	305	417	521
27	68	4800	16 6373	660	3274	648	2528	305	305	255
27	69	5424	15 5877	818	3189	738	2557	610	610	306
28	70	1600	17 6003	4885	3386	584	2527	914	267	605
26	71	4800	15 6019	636	3386	743	2595	914	368	503
33	72	4958	17 166	1443	2593	410	2568	61	61	61
26	73	4800	14 5507	278	3274	687	2649	305	305	138
29	74	5424	15 5877	818	3145	444	2589	305	154	507
29	75	1600	16 48	4958	3326	605	2632	310	194	430
28	76	1600	17 6003	4885	3317	603	2513	610	307	544
26	77	4800	14 5507	278	3274	635	2688	305	155	474
28	78	4800	16 6373	971	3317	604	2652	610	192	240
22	79	1600	12 396	5209	3274	654	2535	305	305	305
29	80	4800	16 114	348	3261	152	2587	61	61	61
26	81	4800	15 6019	220	3274	714	2598	305	305	291
29	82	5314	17 5674	661	3685	798	2523	305	305	152
26	83	4800	15 6019	636	3274	686	2621	305	232	623
26	84	4800	15 5843	278	3317	737	2663	610	378	306
26	85	4800	14 5507	278	3261	619	2646	61	61	61
29	86	1602	16 30	4958	3274	603	2642	305	191	427

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METER READINGS

FLIGHT		AZIMUTH			RANGE			ALTITUDE		
TIME	PATH	INITIAL	TIME X-OVER	FINAL	INITIAL	X-OVER	FINAL	INITIAL	X-OVER	FINAL
26	87	4800	14 5507	278	3261	635	2668	61	153	348
27	88	5424	15 5877	818	3131	421	2540	61	61	61
28	89	4800	16 6382	1397	3261	711	2528	61	479	278
28	90	4800	16 6382	1397	3261	54	2565	61	175	516
28	91	1600	15 801	5270	3261	397	2605	61	292	365
28	92	4800	16 6382	1397	3386	723	2517	914	497	155
27	93	1600	16 30	5743	3261	576	2564	61	61	495
24	94	1328	12 859	5980	2719	709	2535	610	610	362
28	95	4800	16 6382	1397	3386	614	2584	914	318	605
26	96	1600	15 384	5766	3386	743	2595	914	368	503
28	97	4800	16 6373	1232	3386	782	2637	914	533	480
28	98	1600	16 6289	5405	3386	347	2572	914	318	264
29	99	4800	16 114	348	3317	237	2601	610	192	275

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BUILT-IN-TEST FOR STG

PART I

Place the console built-in-test switch to position 1. Depress the following switches on the console in the order shown in table below and observe indications.

STEP	SWITCH	INDICATION
1	"ON/OFF	ON/OFF indicator lights
2	RESET	RESET indicator lights, resets console
3	DEPRESS BORESIGHT (if lit)	BORSIGHT INDICATOR Light Out
3-A	SIMULATOR/TESTER (depress twice)	Both sections should light alternately. When TESTER switch is depressed the AZ meter M3 and RANGE meter M2 should be in green band.
4	Place SIMULATOR/TEST switch in SIMULATOR MODE	
5.	ECM	a. Lights while depressed b. Flashes when released
6	RESET	ECM indicator light out
7	a. RUN/HOLD) (depress twice) b. Place in HOLD before proceeding to Step 8 .	Both sections should light alternately (one always ON)
8	a. RADAR MODE b. MANUAL MODE C. RADAR MODE AND MANUAL MODE	a. Stays ON b. MANUAL MODE on, RADAR MODE off, READY TO FIRE lights c. RADAR MODE: MANUAL MODE and READY TO FIRE light.
9	MANUAL MODE	READY TO FIRE AND MANUAL MODE light.
10	Depress BORESIGHT and adjust AZ and EL pots so tracking error cross hairs are outside of the 60 mil reticle,	BORESIGHT indicator lights.
11	Depress RADIATE	RADIATE indicator lights

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STEP	SWITCH	INDICATION
12	Depress LOCK ON	LOCK ON indicator OFF
13	Depress RADIATE and LOCK ON	RADIATE, LOCKON and ATD light.
14	Depress RADIATE (release LOCK ON) and bring cross hairs on TRACKING ERROR meter Co center of reticle by using AZ and EL adjust-ment knobs.	RADIATE, LOCK ON and ATD indicators light,
15	Depress MANUAL MODE, RADIATE and READY TO FIRE	MANUAL MODE, RADIATE, LOCK ON and ATD indicators light, READY TO FIRE goes out.
16	Depress ATD	Remains out
17		Remains out
18	X over F1/F2	X over F1/F2 and FLIGHT ENDED indicators light.
19	FIRE 1/FIRE 2	Remains out
20	a. Depress MANUAL MODE, X over F1/F2 and FIRE 1	a. X over F1 blinks, MANUAL MODE, FIRE 1, READY TO FIRE AND FLIGHT ENDED indicators light, RADAR MODE goes out.
	b. Depress MANUAL MOI)E, X over F1/F2 and FIRE 2	b. MANUAL MODE, X over F1/F2 blinks, FIRE 2, READY TO FIRE and FLIGHT ENDED indicators light, RADAR MODE indicator goes out.
21	Depress RESET	FIRE l/FIRE 2 indicators out
22	Depress MANUAL MODE, RADIATE, READY TO FIRE, and FIRE 1 and FIRE 2	FIRE 1 and FIRE 2 indicators blink, LOCK ON and ATD indicators light, READY TO FIRE goes out.
23	Depress RESET	
24	a. Depress MANUAL MODE and FIRE 1 and X over F1/F2	a. X over F1/F2, READY TO FIRE and FLIGHT ENDED indicators light
	b. Depress X over F1/F2, and FIRE 2 and MANUAL MODE	b. X over F2 blinks, READY TO FIRE indicator lights.

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STEP	SWITCH	INDICATION
25	Depress RESET	FIRE 1/FIRE 2 indicators out
26	Depress X over F1/F2 and FLIGHT ENDED	X over F1/F2 and FLIGT ENDED indicators light, AZ and EL digital indicators display "88".
27	Depress F1 ERROR/F2 ERROR (depress twice)	Fl ERROR/F2 ERROR indicator should light alternately - one always on.
28	Position AZIMUTH cross hair on TRACKING ERROR meter to left or right extreme of the reticle	AZIMUTH ERROR indicator indicates 49
29	Position ELEVATION cross hair on TRACKING ERROR meter to maximum high or low extreme	ELEVATION ERROR indicate: indicates 49.
30	a. Position AZIMUTH and ELEVATION cross hairs on TRACKING ERROR meter to center of reticle, place F1 ERROR on.	READY TO FIRE and FIRE 1 indicators light, AZIMUTH and ELEVATION ERROR indicators read less than “1O”.
	b. . Depress MANUAL MODE and FIRE 1	
	c. Position AZIMUTH and ELEVATION cross hairs on TRACKING ERROR meter to. maximum extreme outside of reticle	
	d. Depress MANUAL MODE and FIRE 2	FIRE 2 indicator lights
	e. Depress F2 ERROR	AZIMUTH and ELEVATION ERROR indicators read 49.
31	Depress RESET	RESET indicator lights, resets console.
32	Depress BORESIGHT twice	Lights in BORESIGHT MODE out in OPERATE MODE.

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PART 11

STEP	BIT SWITCH POSITION	PROCEDURE	AZIMUTH METER INDICATION
	Pre-test set-up	1. Disconnect interconnect cables from: STG if connected. 2. STG RUN SELECT on #52 (STG Power ON). 3. Out of BORESIGHT (lite OFF) 4. HOLD MODE . 5. RESET 6. Place Simulator/Tester Switch in Test Position	
1	1		Green band
2	2		Green band
3	3		Green band
4	4		Green band
5	5		Green band
6	6		Green band
7	7		Green band
8	1	Depress FLIGHT ENDED switch and AZ ERROR and EL ERROR DIS PLAY 88	
9	3	Procedure: 1. STG Console in HOLD 2. Press RESET 3. STG Console in RUN 4. Press RESET 5.. Observe EL ERROR indicator EL ERROR counts to 24 seconds	

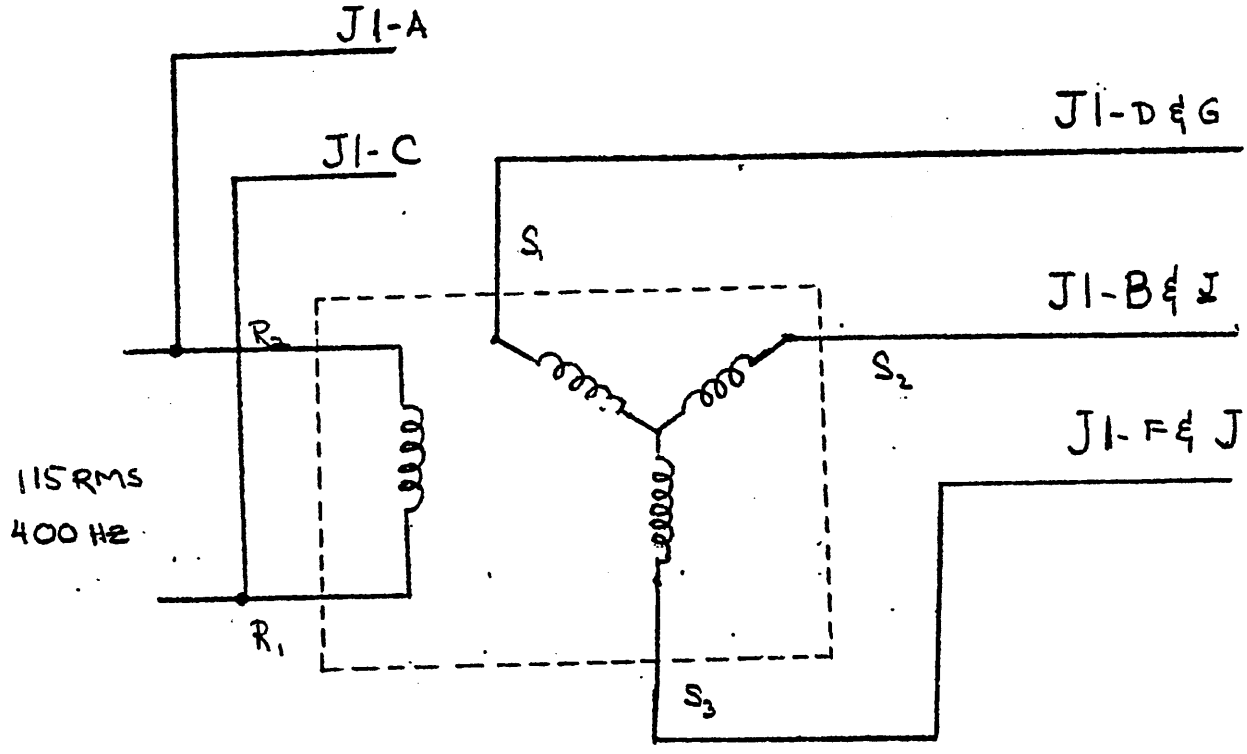
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STEP	BIT SWITCH POSITION	PROCEDURE	AZIMUTH METER INDICATION	RANGE METER INDICATION
10	20		Blue band	Green band
11	4	Procedure: Same as Step 9 EL ERROR counts to 04 seconds		
12	21		Blue band	Green band
13	22		Blue band	Green band
14		Procedure: 1. STG Console in HOLD 2. RESET		
15	8		Green band	Green band
16	9		Green band	Green band
17	10		Green hand	Green band
18	11		Green band	Green band
19	12		Blue band	Green band
20	13		Green band	Green band
21	14		Green band	Green band
22	15		Blue band	Green band
23	2	Procedure: Same as Step 9 EL ERROR counts to 16 seconds		
24	16		Blue band	

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STEP	BIT SWITCH POSITION	PROCEDURE	AZIMUTH METER INDICATION	RANGE METER INDICATION
25		Procedure: Same as Step 14		
26	1 7		Green band	Green ban
27	1 8		Blue bend	Green ban
28	1 9		Green bend	Green ban

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MIL-G-48438A(MU) Generator, Target, Simulated 11741100

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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